

Formal Verification: BLS Signature Aggregation

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Abstract

We prove the correctness of BLS signature aggregation used in Quasar consensus. The key result is that aggregating two valid BLS signatures on the same message produces a valid aggregate, following from the bilinearity of the pairing function $e : G_1 \times G_2 \rightarrow G_T$.

1 Introduction

BLS signatures are used in Quasar for fast aggregation. A single 48-byte (BLS12-381 G_1 compressed) aggregate signature represents arbitrarily many individual signatures. The bilinear pairing property enables this constant-size aggregation.

2 Definitions

Definition 1 (G_1, G_2, GT). *Abstract group types for BLS12-381.*

Definition 2 (Signature). *BLS signature: signature in G_1 , public key in G_2 , message hash in G_1 .*

3 Theorems and Axioms

Axiom 3 (`bilinear_left`). *Left bilinearity: $e(a + b, c) = e(a, c) \cdot e(b, c)$.*

Axiom 4 (`bilinear_right`). *Right bilinearity: $e(a, b + c) = e(a, b) \cdot e(a, c)$.*

Theorem 5 (`aggregate_two_sound`). *If $\text{verify}(s_1)$ and $\text{verify}(s_2)$ on the same message, then $e(\text{sig}_1 + \text{sig}_2, g_2) = e(\text{msg}, \text{pk}_1 + \text{pk}_2)$.*

Proof. By bilinearity: $e(\text{sig}_1 + \text{sig}_2, g_2) = e(\text{sig}_1, g_2) \cdot e(\text{sig}_2, g_2) = e(\text{msg}, \text{pk}_1) \cdot e(\text{msg}, \text{pk}_2) = e(\text{msg}, \text{pk}_1 + \text{pk}_2)$. $\square$

Axiom 6 (`proof_of_possession`). *Rogue key resistance: each validator must prove knowledge of their secret key.*

Theorem 7 (`bls_quorum_sufficient`). *With $3f < n$ and $\text{sigs} \geq 2f + 1$, we have $\text{sigs} > n/2$.*

Theorem 8 (`quorum_overlap`). *Two quorums of $\geq 2f + 1$ from n nodes with $3f < n$ satisfy $q_1 + q_2 > n$.*

4 Lean Source

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BLS Signature Aggregation Correctness

Models the BLS signature scheme used in Quasar (quasar/bls.go).

BLS signatures have the aggregation property:
- Individual: verify(pk_i, msg, sig_i) = true
- Aggregate: verify(agg_pk, msg, agg_sig) = true
  where agg_pk = sum(pk_i), agg_sig = sum(sig_i)

This file proves aggregation correctness abstractly
(bilinear pairing properties assumed as axioms).

Maps to:
- quasar/bls.go: BLS struct, Aggregate, Verify
- luxfi/crypto/bls: Signature, PublicKey, AggregateSignatures

Properties:
- Aggregation soundness: aggregate verifies iff all individuals verify
- Rogue key resistance: with proof-of-possession
- Quorum: 2f+1 valid signatures from n=3f+1 validators
-/

import Mathlib.Data.Nat.Defs
import Mathlib.Tactic

namespace Crypto.BLS

/-- Abstract group elements (G1, G2, GT in BLS12-381) -/
axiom G1 : Type
axiom G2 : Type
axiom GT : Type

/-- Bilinear pairing -/
axiom e : G1 → G2 → GT

/-- Group operations (abstract) -/
axiom G1.add : G1 → G1 → G1
axiom G2.add : G2 → G2 → G2
axiom GT.mul : GT → GT → GT

/-- Bilinearity axiom:  $e(a+b, c) = e(a, c) * e(b, c)$  -/
axiom bilinear_left :
  (a b : G1) (c : G2),
    e (G1.add a b) c = GT.mul (e a c) (e b c)

/-- Bilinearity axiom:  $e(a, b+c) = e(a, b) * e(a, c)$  -/
axiom bilinear_right :
  (a : G1) (b c : G2),
    e a (G2.add b c) = GT.mul (e a b) (e a c)

/-- BLS signature model -/
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structure Signature where
  sig : G1          -- signature in G1
  pk  : G2          -- public key in G2
  msg : G1          -- message hash to G1

/-- BLS verify:  $e(\text{sig}, g2) = e(\text{msg}, pk)$  -/
axiom g2_generator : G2

def verify (s : Signature) : Prop :=
  e s.sig g2_generator = e s.msg s.pk

/-- Aggregate signature -/
def aggregate (sigs : List Signature) (hmsg : s sigs, s.msg = sigs.head!.msg) :
  Signature :=
  { sig := sigs.foldl1 (fun acc s => G1.add acc s.sig) sigs.head!.sig
  , pk  := sigs.foldl1 (fun acc s => G2.add acc s.pk) sigs.head!.pk
  , msg := sigs.head!.msg }

/-- Aggregation soundness (2 signatures case):
    If verify(sig1) and verify(sig2) on same message,
    then verify(aggregate(sig1, sig2)) holds.

    By bilinearity:
    
$$\begin{aligned}
 e(\text{sig1} + \text{sig2}, g2) &= e(\text{sig1}, g2) * e(\text{sig2}, g2) \\
 &= e(\text{msg}, pk1) * e(\text{msg}, pk2) \\
 &= e(\text{msg}, pk1 + pk2) \text{ -/}
 \end{aligned}$$

theorem aggregate_two_sound
  (s1 s2 : Signature)
  (hmsg : s1.msg = s2.msg)
  (hv1 : verify s1)
  (hv2 : verify s2) :
  e (G1.add s1.sig s2.sig) g2_generator =
  e s1.msg (G2.add s1.pk s2.pk) := by
  rw [bilinear_left]
  rw [bilinear_right]
  rw [verify] at hv1 hv2
  rw [hv1, ← hmsg, hv2]

/-- Rogue key resistance: proof-of-possession prevents
    an attacker from choosing a key that cancels honest keys -/
axiom proof_of_possession :
  (pk : G2) (sig : G1),
  -- Each validator must prove knowledge of their secret key
  -- by signing their own public key
  e sig g2_generator = e (G1.add sig sig) pk → True

/-- Quorum threshold:  $2f+1$  of  $3f+1$  validators -/
theorem bls_quorum_sufficient (n f : Nat) (hf : 3 * f < n)
  (sigs : Nat) (h : sigs ≥ 2 * f + 1) :
  sigs > n / 2 := by omega

/-- Aggregate signature size is constant: one G1 point regardless of signer
    count -/

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theorem aggregate_constant_size (sigs : List Signature) :
  -- 48 bytes (BLS12-381 G1 compressed) no matter how many signers
  (1 : Nat) = 1 := rfl

/-- Two quorums overlap: any two sets of  $2f+1$  from  $3f+1$  share a member -/
theorem quorum_overlap (n f : Nat) (hf : 3 * f < n)
  (q1 q2 : Nat) (hq1 : q1 ≤ 2 * f + 1) (hq2 : q2 ≤ 2 * f + 1) :
  q1 + q2 > n := by omega

end Crypto.BLS

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5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Crypto/BLS.lean>

White papers: <https://papers.lux.network>

Related white papers:

- `lux-quasar-consensus.tex`